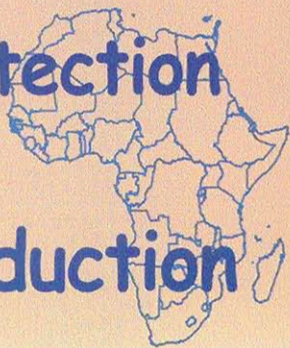


Environmental Protection & Disaster Risk Reduction



A School Guide



Environmental Protection & Disaster Risk Reduction

A School's Guide



International Strategy
ISDR
for Disaster Reduction

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FOREWORD

**We should strive to work in harmony – not destroy –
our environment.**

Indeed, our environment is a most precious resource that can reduce significantly the impact of disasters. Deforestation and soil erosion are just some causes of environmental degradation, and environmental degradation can lead to disasters.

The good news, however, is that protecting the environment can also be done by school students.

This booklet, *A School's Guide to Environmental Protection & Disaster Risk Reduction*, tells us what can be done to protect the environment and in turn reduce the impact of disasters.

It also provides definitions and explanations in school-friendly language, supported by success stories providing inspiration for others to contribute to disaster risk reduction by protecting their local environment.

A stylized, handwritten signature in black ink, appearing to read 'Sálvano', followed by a vertical line.

Sálvano Briceño

Director

United Nations Inter-Agency Secretariat of
the International Strategy for Disaster Reduction

1. INTRODUCTION

The impact of natural disasters in terms of human and economic losses has risen in recent years, and society in general has become more vulnerable to natural disasters. Those usually most affected by natural and other disasters are the poor and socially disadvantaged groups in developing countries as they are least equipped to cope with them.

Affirmation: Yokohama Strategy and
Plan of Action for a Safer World



Floods (Photo: Care Canada)

Have you ever experienced a flood, a landslide, a drought, a cyclone or a wildfire? What about an earthquake or a volcanic eruption? No? Then you are very lucky, because thousands of people throughout Africa experience one or more of these natural disasters every year - and scientists have warned us to expect more frequent and more severe climate-related disasters to happen because of Global Climate Change.

Can natural disasters be prevented? No, but we can protect ourselves by reducing, or better still, eliminating, the factors that put us at risk. This process is called Disaster Risk Reduction (DRR), and has as its central focus, Environmental Protection because it has been shown that the more we damage our environment through pollution and bad land practices, the more we put ourselves at risk of a disaster occurring.

However, because poverty is frequently the root cause of environmental degradation, disaster risk reduction and environmental protection must form part of a broader programme to uplift communities through Sustainable Development (working in harmony with the environment so as to become self-sufficient).

This booklet is aimed at helping us to identify the hazards that put us at risk, suggest ways to reverse or reduce that risk through working in harmony with the environment, and to establish monitoring and early warning systems that will alert us to impending disasters. Remember, we all have an important part to play in preserving our environment so that our communities can thrive.

1.1. A WORD ON DISASTERS

Although we call droughts, floods, earthquakes and wildfires 'disasters' because of the devastating impact they have on communities, they are not in themselves disasters, but are instead natural hazards – which have the potential to become disasters, especially when they strike unprepared communities or environmentally degraded areas.

It is important to understand this distinction because effective disaster management depends on the ability to identify hazards within the area, evaluate the risk attached and then find ways to avoid or at least reduce their potential for creating a disaster.

1.2. HAZARDS AND DISASTERS IN AFRICA

Before examining the hazards within Africa in more detail, let's take a look at some of the recent disasters that have occurred in our Continent. Unfortunately, over the past 30 years, Africa has experienced more and more devastating disasters. In the 1970s and 1980s drought was the main concern, but since the mid 1990s, floods and tropical cyclones have been added to Africa's burdens.

Drought: In January 2004, parts of South Africa were declared disaster areas as a result of what some are calling the worst drought in 88 years. About 14 million people are affected, and the impact on the economy is severe. Extreme drought has also created major food emergencies in Burkina Faso, Chad, Ethiopia, Kenya, Niger, Rwanda, Somalia, Sudan, Tanzania and Uganda (FAO, 2001).

In February 2004, a national disaster was declared in Swaziland because of widespread drought which, the World Food Programme claims, has left about 200,000 people without food. Similarly, Lesotho has also declared an emergency, with some 800 000 people in need of help.

Floods: In 2001, Mozambique was hit by Cyclone Dera, which left 79 000 ha of planted land underwater and nearly 120 000 farm families without food. Flooding also affected the livelihood of about 2 000 families from the fishing community.

In July 2003 more than 25 000 villagers were severely affected by flooding in eastern Caprivi (Namibia), reportedly the worst in decades.

In the first quarter of 2004, heavy rains and flooding affected parts of Angola, Zimbabwe, Zambia, Mozambique and Botswana.

Earthquakes: In North Africa (Egypt and Algeria), 22 earthquakes killed 14 405 people and affected another 106 150 people between 1980 and 1998. In January 2002, lava flowing from Mount Nyiragongo destroyed half of the city of Goma in the eastern Democratic Republic of Congo. More than 400 000 people fled to take refuge in neighbouring villages and in Rwanda.

In 1986, explosive emissions of toxic gases from Lake Nyos and Lake Mounoun, both volcanic crater lakes in the mountainous west of Cameroon, killed thousands of people.

Did you know?

Nearly half of Africa's population is threatened by desertification. At the UN Convention to Combat Desertification it was reported that this threat, when combined with extreme poverty and severe land degradation, will acutely affect millions of people.

(Water Sewage & Effluent Sept 2002 Vol 22 No 4).

2. HAZARDS AND DISASTERS

2.1. WHAT ARE THE HAZARDS?

In Africa, there are four main or primary hazards which have the potential to kill and destroy. These have their origin in the continent's geology, climatic patterns, health issues and irresponsible technological practices.

In many cases, however, secondary hazards become a greater threat to the community than the original primary hazards. For example, tropical cyclones along the east coast of Africa can trigger storm surges, flash coastal and river floods and landslides. Similarly, the landslides, fires, tsunamis and floods that occur after an earthquake often cause more damage than the earthquake itself. In the urban environment, pollution, overcrowding and limited or even no sanitary and medical services can lead to epidemics, chronic illness and an on-going cycle of crime and poverty.

Although we have little control over hazards arising from the geography of Africa, there is a great deal we can do to limit the damage they can do - by simply looking after the environment.

2.2. THE PRIMARY HAZARDS

Geological Hazards

The Earth's crust is made up of tectonic plates which are slowly moving all the time. Where these plates collide or converge, one plate can sink under another forcing the other upwards, sometimes causing folds, fractures or faults in the Earth's crust. These areas then become more at risk from earthquakes, volcanic activity, landslides, rock falls, subsidence and sinkholes. The largest such area in Africa is the East African Rift Valley.

A rift valley is a steep-sided, flat-bottomed valley which is formed by the land lying between two faults subsiding (sinking). The East African Rift Valley is part of a rift system that spans nearly 6 400 km, from the valley of the River Jordan and the Dead Sea, through the Gulf of Aqaba, across East Africa to the coast near Beira in Mozambique. Regions lying within this fault zone are most at risk from geological



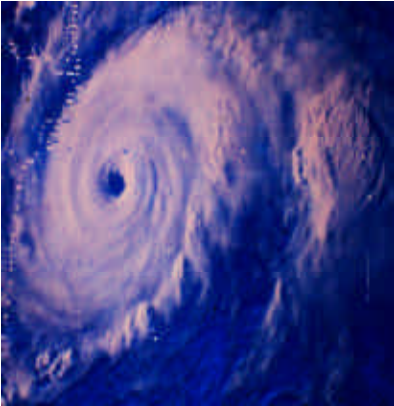
Volcano

hazards such as earthquakes and volcanic activity. When seismically active areas also have a high urban growth rate - as do cities in North and East Africa as well as in the Maghreb region of North West Africa, the potential for earthquake-induced disasters is very high because of collapsing buildings and damage to infrastructure.

Climate-Related Hazards

Much of Africa lies in the path of many extreme weather events, especially torrential rain, cyclones and heat waves. These carry with them the risks of flooding or drought - both of which have the potential to destroy life and property on a large scale. Famine becomes a real possibility because drought generally prevents crop production whilst flooding can cause widespread crop losses.

Scientists have warned that global climate change may well worsen the situation for Africa's people, because the predicted increase in frequency and intensity of these extreme weather events could give rise to prolonged droughts, desertification, flooding and a rise in sea level, all of which pose a threat to local communities. Deforestation, inappropriate coastal development and poor land management will further aggravate the effects of climate variability and change.



Satellite image of a cyclone moving towards the continent of Africa
(Photo: NOAA)



Flood (Photo: IFAD)

Biological Hazards

Biological hazards in Africa are responsible for a significant number of deaths and misery. These include diseases such as HIV/AIDS and Tuberculosis, parasitic infections in the form of Malaria and Bilharzia, while animal sicknesses like Foot and Mouth disease have resulted in huge losses of livestock.



Anophele (Malaria mosquito, photo: WHO)

The extensive infestation of the natural environment by alien plants, introduced from other parts of the world, poses a significant threat to livelihoods in many parts of Africa.



Alien plants pose a significant threat to livelihoods (Photo: GTZ)

Technological Hazards

These are hazards caused by contamination from poisonous or harmful waste matter which has affected the air, water or environment in some way. Some pollutants come from nature – for example toxic gases escaping from fissures in the earth or from volcanoes. However, most technological hazards are created by human activities that threaten the delicate balance of nature that sustains life on Earth – global warming, damage to the ozone layer and acid rain are three examples of man-made hazards. Think of the total disruption of whole communities and the health issues caused by the nuclear disaster at Chernobyl in 1986. The oil spills from pipelines and the tankers that circle our coasts, factories that illegally release poisonous gases into the air or discharge poisonous chemicals or metals into the rivers, or sewage that is not properly treated – and you will understand why it is so important to handle technology responsibly. For example, in Tanzania the National Environment Management Council has warned that most of the country's water supply will become dangerously toxic unless

drastic anti-pollution measures are taken, especially in towns where industries are located.

Today there are many laws aimed at protecting us from technological hazards, but accidents still happen. Often developing countries simply do not have the means to ensure compliance with the law or the ability to deal with the accidents when they occur. In Africa, pollution and health hazards are a constant threat to people who have access to only limited or dysfunctional services, such as those living in the densely populated but under-developed urban areas or in overcrowded informal settlements.



The Niger Delta suffering from environmental pollution (Photo: GTZ)

2.3. WHAT CAN WE DO?

Firstly, we must try and reduce the chances of a disaster happening; the best way to do this is to care for our environment - the trees, soil, vegetation, water supply, atmosphere - because it in turn helps to keep us healthy and is essential to the development of our communities.

Secondly, we must try and limit the amount of damage disasters can do; this is achieved by taking proactive steps to limit the impact, by



Community cares of their environment

putting early warning systems into place and by making sure in advance that everyone knows what to do in the event of an emergency.

Disaster Risk Reduction looks for ways to achieve these two objectives, because disasters have a very high economic and social cost. International development banks (including the World Bank) and agencies support the strengthening of national commitments to engage in disaster risk reduction activities.

It is important that all development projects include a critical consideration of potential disasters and that suitable disaster risk reduction measures are included in environmental protection and poverty reduction programmes.

3. DISASTER RISK REDUCTION

Communities must adopt the notion that disaster impacts can be reduced and therefore not only wait for disasters to be managed. In some cases, it might be possible to reduce hazards themselves. If not, then it would certainly be possible to reduce human vulnerability to those hazards.

ISDR 2002



Stagant water can be harmful to your health (Photo: GTZ)

3.1. HOW CAN WE REDUCE DISASTERS?

Is your community at risk of a disaster? And how can we determine the severity of the risk?

The United Nation's International Strategy for Disaster Reduction (ISDR) uses the following formula:

$$\text{Risk (R)} = \text{Hazard (H)} \times \text{Vulnerability (V)} / \text{Capacity (C)}$$

In simple terms, this means that the risk of a disaster happening depends on what the hazards are, how vulnerable the community is, and how well they can limit the damage by being prepared and

forewarned. The risk of a disaster occurring is based on physical, economic and environmental factors, which need to be monitored and evaluated continuously.

Disaster Risk Reduction is everyone's responsibility and involves many areas of activity. Here we focus on:

1. risk analysis and assessment;
2. developing an awareness of risk;
3. developing early warning systems; and
4. disaster risk management.

Risk Analysis and Assessment

Step 1 in assessing risk is to identify the hazards involved. For example, does the area lie within a cyclone or seismic belt; is there any evidence of air or water pollution, or land degradation? Is the village close to a river that has seasonal flooding?

Scientists are able to provide this information by monitoring key indicators and collecting geographical, atmospheric and environmental data. This information is then mapped and analysed in order to develop a clearer picture of the hazards – and therefore of the risks – involved.

It is also important to know whether the activities and attitudes of the local people are adding to the risk of disaster. For example, are they degrading their environment, do they have the means to cope with a disaster, do they recognise that their community could be at risk, and if so, have they developed an early warning system and disaster management plan? A programme aimed at reducing the risk of disaster can be designed around the answers to most of these questions.

Developing Risk Awareness

If a community does not see itself to be at risk, it is not likely to actively participate in disaster risk reduction programmes. Developing risk awareness begins with understanding how the local people see their world and determining what for them would be an acceptable level of risk.



Risk Mapping (Photo: Chip Snaddon)

It is easy to recognise the dangers of rising rivers, desertification, grumbling volcanoes and polluted water; it is less easy to see that global climate change, damage to the environment - such as deforestation, soil erosion and loss of biodiversity - can be just as dangerous. If the local community is actively involved in the risk assessment process, it is an opportunity to raise their awareness about potential hazards. Some of these hazards may be familiar, others not. However, once a hazard is recognized it is possible to effect a positive change in the people's response and this contributes towards disaster risk reduction.

Having created community awareness, the next stage is a commitment from the local public authorities to implement disaster risk reduction policies. To do this, it may be necessary to re-allocate funds from visible development projects - such as building houses - to less obvious projects - such as soil conservation - that are aimed at limiting the impact of longer-term threats. It is important that the community understands the reasons for this.



Use of wood for energy (Photo: Care Canada)

Developing Early Warning Systems

Advance warning of an impending disaster gives the community time to try and protect itself - for example, to take preventive measures to reduce potential impacts of disaster and to evacuate to higher ground in the case of a flood risk - and by so doing, reduce the possibility of injury, loss of life and damage to property. Early warning of slow-onset disasters such as drought and famine is equally important as it gives the community time to find ways to counteract the threat and introduce community support measures.

Developing an early warning system is therefore a vital part of disaster risk reduction. This should be done at a community level and includes the following three key steps:

- **Forecast and Prediction** - just as it is possible to predict the weather by collecting information about the atmosphere on an on-going basis, it is possible to predict a slow-onset or rapid-onset disaster by collecting and monitoring information gathered from the environment. We need to be able to read and understand the warning signals a stressed environment gives out.
- **Announcing the Warning** - the next step is for community leaders to decide on an effective way of warning both the local people and neighbouring communities, and if necessary, the regional government of the threat and its possible impact on health, livelihood and infrastructure. For example, when heavy rains caused landslides in Kenya's Central Province in May 2004, the instruction to move villagers at risk to higher ground passed from the Commissioner to the Chiefs to the Village Headmen.

The community should also be encouraged to look at its land-use practices or way of life to make sure that they understand what practices increase the risk of disaster and what activities will reduce that risk. These proactive steps are called **provention**.

- **Reaction** - once community leaders and villagers fully understand and accept the information and advice received, they must be encouraged to implement the necessary prevention or protective measures.

Disaster Risk Management

Each country bears the primary responsibility for protecting its own people, infrastructure and other natural assets from the impacts of natural disasters.

(Principle 9 of the Yokohama Strategy and Plan of Action for a Safer World)

Until recently, disaster management focused mainly on providing emergency relief to communities struck by disaster. Today, the approach has broadened to include programmes for empowering people to help themselves.

As we have seen, once people can recognise the factors putting them at risk and know what to do to minimize that risk, the more chance they have of reducing the likelihood of a disaster occurring at all, or at the very least, limiting the extent of the damage if it does. This is particularly true of slow-onset disasters because many of them - desertification, famine, flooding, wild fire, disease and epidemics - arise from environmental degradation which is largely avoidable.

A prepared community is a resilient community that is better able to withstand and recover from a disaster. Take the case of a volcano or earthquake; in the immediate aftermath of a disaster, the affected



Disaster preparedness (Photo: IFRC)

community is in a state of shock and is often not able to think beyond its immediate needs. On the other hand, a community that has strong community leadership and has prepared in advance for a disaster knows immediately what it needs and how to respond. It is also able to play an active part in directing relief operations and initiating important medium and long-term developments during the recovery phase.

Without this preparedness and resilience a community can become dependent on outside assistance and reliance upon international relief agencies can become entrenched.



Community well prepared for disaster (Photo: IFRC)

4. THE BENEFITS OF ENVIRONMENTAL PROTECTION

Environmental management tools have the potential to "...make a substantial and cost-effective contribution to reducing the vulnerability to natural hazards". (ISDR 2002)

We have seen that the risk of disasters occurring in Africa is high because of the geological, climatic, technological and health hazards that exist here. We have learnt that whilst some disasters are not preventable, others are – especially slow-onset ones such as famine. We have also established that the impact of all disasters can be reduced if communities have effective disaster risk reduction practices in place.

Now we are going to look at why environmental protection and the sustainable use of natural resources are so important in disaster risk reduction measures.

4.1. WHY SHOULD WE PROTECT THE ENVIRONMENT?

The 'environment' is made up of the earth, water, atmosphere and biosphere and is influenced by the climate, the soils, the topography and human activities. When the environment becomes degraded as a result of natural hazards or our activities, all forms of life become threatened because natural processes and or ecosystems are altered or damaged permanently. As a result, the environment becomes more vulnerable to natural hazards, which may in turn become more frequent and intense.

Loss of biodiversity results in an increasingly unstable environment and eventual destruction of ecosystem functioning. This impacts on the quality of the goods and services that the ecosystem delivers and also adversely affects a community's livelihood options. This in turn increases an area's dependence on man-made infrastructure and supplies from outside.

Causes and effects of environmental degradation

Deforestation: This results in the loss of wood for cooking (energy) and building materials, plants and animals (for food and medicine) and also exposes the land to the risk of soil erosion.

Land degradation: Soil erosion reduces the quality and quantity of the soil and increases desertification. Other impacts are declining agricultural yields, increased risk of flooding, increased competition for resources and potential for conflict.

Air pollution: Concentrations of population, industry and vehicles in urban areas and increase in wood burning fires in rural areas is increasing air pollution, which in turn increases the health risk and risk of food shortage.

Desertification: This impacts on food production and increases the migration off the land to urban areas where many people live in unhealthy and poor conditions without the necessary skills to earn a good living.

Coastal degradation: This results in vegetation loss and soil erosion and increases the vulnerability to devastating damage to property and loss of life during high seas and cyclones.



Poor water resource development and management: This can result in high algal growth in lakes and dams, contamination of groundwater and the loss of aquatic habitats and biodiversity. These constrain food production and industrial activities, and contribute significantly to the burden of disease, and the risk of famine.



Informal settlements: These are often closely linked to poverty and environmental degradation in rural areas. Because there are not enough houses and basic services available in the urban areas, large informal settlements develop around the bigger towns and cities, and the urban environment becomes degraded as a result. The risk of epidemic and grinding poverty is increased.



Loss of biodiversity: Plants and animals comprise a natural "community" and the conditions define a habitat. If a habitat is degraded, fewer kinds or species of plants and animals can thrive in it and biodiversity is lost.



Projects aimed at reducing the risk and consequences of disasters through environmental protection are more effective if they are community-driven at a local, regional and even international level. For example, in Niger, community involvement has been key in preventing uncontrolled deforestation. Niger is about 80% desert, but nearly 95% of its rapidly growing population is dependent upon its small wooded area for fuel. If the forest and its ecosystem is to survive, wood will have to be harvested in a sustainable way. To help solve the problem, the authorities have given local communities exclusive tree-cutting rights over specific forest areas and established rural wood

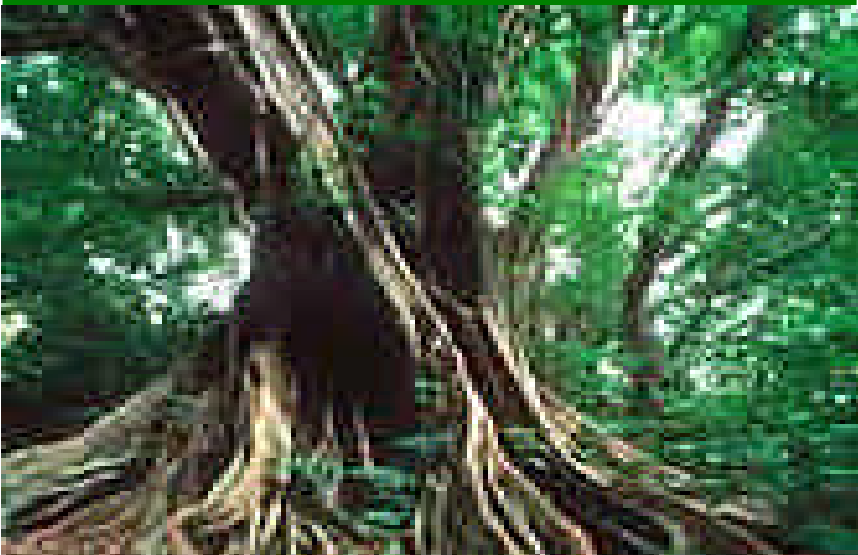
Did you know?

About 17% of the forests in the world are found in Africa. These forests are concentrated mainly in the tropical zones of western, central, eastern and southern Africa. However, large tracts of intact natural forest are now found only in Central Africa.

The forests of the Congo Basin are second in size to the Amazon rainforest, and are home to three of the world's four species of Great Apes (gorillas, chimpanzees and bonobos), half the elephants of Africa as well as rare animals such as the okapi and bongo. Sadly, these animals risk extinction because of poachers, and large areas of forest have been cleared by irresponsible loggers and a growing population in need of land.

Six African countries and several western nations have formed the Congo Basin Forest partnership with a view to stopping illegal logging and finding a way for the local people to benefit from the forest without destroying it.

Vast areas of the forests in West Africa have been chopped down - mainly for fuel.



Congo Basin's forests could be lost within fifty years if illegal logging, poaching and smuggling of wildlife, and illicit bushmeat trade continue at current levels.

markets - managed by those communities - at which wood can be sold. To enable the forest to regenerate, only selected species of tree are allowed to be cut and then only at certain times of the year and in specified amounts.

Thus, Environmental Protection encourages people to find creative ways to use their indigenous skills so as to sustain themselves without degrading the environment or becoming dependent on outside support. It illustrates the paradox that learning new methods and adapting old ways can be a key to securing the future of indigenous culture.

The value of Madagascar's forests

In Masoala National Park the income generated from forest products sustainably harvested by local villagers over ten years could be as much as US\$200 000. Income earned using slash and burn agriculture over ten years was estimated at US\$12 000 with severe environmental degradation. Selling the timber on the global market using commercial forestry would realise income of US\$90 Million over ten years but also the ultimate loss of the forest. It is not sustainable and has global climate impacts. Preserving the forest for purposes of global climate regulation would save more than US\$180 million and produce acceptable income for local village.

Niger River Basin management

Nigeria, Benin, Burkina Faso, Cameroon, Mali, Chad, Guinea, Ivory Coast and Niger have agreed to jointly oversee the development and management of the Niger River basin, which is the third biggest watercourse in Africa. About 110million people rely on it for fresh water, fishing and agriculture.

4.2. WE EACH HAVE A ROLE IN ENVIRONMENTAL PROTECTION

The concept of "Ubuntu" is understood throughout most of Africa. It means that one is not a full human being if another human being is suffering. It is a philosophy that holds families and communities together and encourages the sharing of resources for the benefit of all.

Disaster risk reduction and environmental protection relies on this collective responsibility and the success stories given later in this booklet clearly illustrate the role that responsible leaders can play in uplifting their communities. These stories also illustrate the extraordinary perseverance and talent of women who have initiated and continue to lead environmental protection measures. The result has been to give ordinary people new skills, a sense of hope and the ability to actively participate in shaping their future.



Community should look for another alternative for energy (Photo: IRIN)

4.3. COMMUNITIES

Communities are often closely knit and bound by common cultures, family ties and traditional ways. Thriving communities are those that have learnt how to live in harmony with themselves and their environment. To achieve this, the community may have to learn new ways of overcoming old problems, but this can only come about if its

members understand the consequences and cost of over-exploiting their land and overstressing their ecosystems.



Training and educating communities in fighting bush fires is sound risk management practice (Photo: Wouter van Warmelo).

4.4. THE CONCEPT OF SUSTAINABLE DEVELOPMENT

An area with a healthy and diverse ecosystem is better able to absorb the impact of a natural disaster than a degraded environment and it is one that can recover more easily. We therefore need to ensure that our activities do not damage the diversity and integrity of nature. This means we must harvest resources in a sustainable manner and interact with our environment in a way that does not lead to land degradation, desertification, deforestation, loss of biodiversity, land, water and air pollution or wildfires. Because many of these destructive practices have been caused by poverty, it is important to find ways in which the local population can earn a livelihood without having to degrade the environment. This is the essence of sustainable development, which, if properly undertaken, will lead to an improvement in the quality of life for the present generation and ensure that future generations also enjoy the benefits of a healthy environment. Disaster risk reduction and environmental protection support and complement each other since both aim to promote sustainable development.

Three important components of sustainable development are:

- The Social Capital - which consists of the people and their skills as well as the systems of democratic governance, social services and the quality of government departments and other community organizations.
- Growth in Economic Capital which consists of the national, regional and local infrastructure such as buildings, roads, ports, railways, water supply schemes as well as the Financial Capital or money available.
- Environmental protection or safeguarding the Natural Capital which consists of all the resources provided by the earth such as soil, water, plants, forests, wildlife, fish, minerals, oil and gas.

A community's wealth is measured by its total capital - the sum of its social, economic and natural capital. For development to be sustainable, this total capital should increase with time.



Our environment can be our wealth (Photo IFAD)

Examples of natural capital provided by different environments

Natural Systems	Good	Services provided
Wetlands, e.g., Okavango Swamps.	Water, food and useful plants.	Help to purify water, act as a sponge to store water and reduce the risk of flooding.
Coastal dunes and mangroves, e.g., the coast of Nigeria.	Wood	Protect the coast during storms; Ecotourism opportunities.
Natural vegetation, e.g., Savannah and Forest of Kenya, Tanzania, Congo Uganda.	Food (fruit, berries, wild animals) Raw materials (wood) Building materials (wood, thatch, clay, stone Medicinal plants	Stabilizes the soil and prevents soil erosion; Forests produce oxygen and absorb carbon dioxide for the continent and the planet; Ecotourism opportunities
Mountains	Water, useful plants	Act as catchments, storage areas for water Ecotourism opportunities

Disasters can also lead to a loss of social, economic and natural capital. Repeated exposures to disasters from which a community is less and less able to recover will lead to a downward spiral of poverty. Adopting disaster risk reduction measures can prevent or reverse this negative cycle.

4.5. SUCCESS STORIES

The World Wildlife Fund's involvement in Namibia

The Community Game Guards project, initiated by WWF-South Africa, is a new approach to wildlife conservation based on the grounds that the local indigenous people, with training, make the best conservationists. This project and other similar community-based projects, has been so successful that many species of wildlife have returned to areas of Namibia from which they had disappeared as a result of hunting, illegal poaching and harassment by local people.

Due to the success of the Community Game Guards project, others have been initiated; one is The Integrated Rural Development and Nature Conservation Project with the objective of improving the effectiveness of the community game guard networks in the Kunene and Caprivi regions.



Game guard tracking rhinos, Torra Conservancy, Kunene, Namibia (Photo: WWF)

Cameroon woman wins major international award for disaster reduction

Mrs Tadzong of Cameroon won the prestigious 2003 Sasakawa Award for Disaster Reduction because of her long-term personal commitment and dedication to disaster risk reduction. Together with the Global Centre for Compliance, Hazards and Disaster Management, Mrs Tadzong has promoted the use of environment-friendly practices that mitigate disaster risks at community level. As a result, Mrs Tadzong and her colleagues have played an important role in helping the poor to sustain their safety and livelihood through their educational, social and economic activities at the community level. She strongly believes that one can reduce the impacts of disasters if preventative steps and actions are taken before they occur.



Mrs. Tadzong (In the middle) and her colleagues

Their projects have included:

- Community-based disaster prevention training (e.g. advising people not to hide under trees or steep slopes when it rains, because of landslide, soil creep and rock fall);
- Advise and support to small farmers to integrate small trees into crop land to maintain soil fertility and prevent land degradation;

- Reduction of hazards through waste management and compost production.

She has succeeded in combining disaster risk reduction, environmental protection and agricultural activities. This award is a great honour for all African women who play such an important role in maintaining the well being of communities before, during and after disasters.

Working for Water- South Africa

Working for Water (WfW) is a South African programme, which simultaneously tackles the problems of invading alien plants and unemployment. It is a national government programme to:

- Enhance water security, improve ecological integrity, restore the agricultural potential of land and promote sustainable use of the natural resources;
- Invest in the poor sectors of South African society.



Women are keenly involved in the Working for water alien vegetation clearing programme (Photo: Wouter van Warmelo)

In the national parks of South Africa, the single biggest threat to biodiversity comes from invading alien organisms. The WfW programme, together with South African National Parks, works to control invading alien plants, while at the same time stimulating a process of job creation and social transformation in the communities close to the parks.

SOS Sahel community forestry project in Ed Debba, Sudan

On the banks of the Nile, near Ed Debba, Sudan, there are numerous environmental problems. These include the encroachment of sand into productive land, uncontrolled tree cutting and overgrazing, and the poor living conditions of the rapidly increasing population. The SOS Sahel Community Forestry Project has successfully addressed many of these problems.

- Sand encroachment has been reduced by establishing vegetated belts and windbreaks and stabilising the dunes;
- Tree cutting and overgrazing have been reduced by harvesting seed pods from trees for fodder and fuel;
- Poor living conditions have been improved by employing women in home-based activities, protecting houses, water points and infrastructure and constructing better houses.



Water point in a rural zone (Photo: IRIN)

Improved community environmental protection has resulted in better living conditions and the community has reduced the risk of having to abandon their homes to relocate elsewhere. Other villages in the Nile basin are starting to follow these important advances.

Urban upgrade in Senegal

In the small Senegalese town of Rufisque, a lack of sanitation posed a health risk. Much of the residential land lies below sea level and the ground water sources of drinking water are easily polluted by sewerage from the pit latrines. Through community participation, aided by International Funding and the Rufisque Local Authority, nine low-income communities are well on the way to solving the problem by using horse drawn carts to collect the solid waste and low-cost pipes dispose of wastewater and sewerage. Sewerage, wastewater and solid waste all end up in purification and recycling centre where it is treated to form compost for use in the local market gardens.

The local community actively participates in this scheme and women play a prominent role. The safe disposal of solid waste, the elimination of excrement as a source of disease and the reduction of flies and mosquitoes have all improved the urban environment and reduced health risks to the community.



One of urban town in Senegal (Photo IRIN)

4.6. INTERNATIONAL BENEFITS AND RESPONSIBILITIES

The international community has benefited from the use of Africa's resources such as minerals, oil and gas deposits, timber, agricultural products and its unique tourism sites. In return it has accepted that it has a responsibility to help protect Africa's natural capital in co-operation with the local communities. Most foreign states also recognise that poverty reduction, the vulnerability of people to natural disasters and environmental protection are linked and this forms an important part of the UN Millennium Declaration.

The people of Africa are the custodians of their continent and have a responsibility to work with each other and the international community to actively protect Africa's natural environment to ensure the sustainable development of its non-renewable resources.

Good examples of this approach are the projects managed under the Ramsar Convention which is a global environmental treaty whose mission is the "conservation and wise use of Wetlands by national action and international cooperation".



The beauty of wetlands (Photo: Ramsar)

Examples of Ramsar Projects in Africa

Ramsar Project	Environmental Protection	Disaster Risk Reduction Benefit
Niger River Basin	Protect wetlands	Reduce flood risk and improve water quality
Oasis de Tamantit et Sid Ahmed Timmi, Algeria	Protect unique desert oasis ecosystem	Primary source of water supply, safeguarding against famine and drought; it is used in religious activities and is a tourism destination
Lake Chad, Niger	Protect lake habitats and biodiversity	Preserve water supply, reduce desertification and famine. The site is used for fishing, salt production and harvesting of Spirulina, hunting, small-scale agriculture and extensive grazing of sheep and camels
St Lucia System, South Africa	Protect wetlands and biodiversity	Used for ecotourism, education, and in religious activities
Complexe des lacs de Manambolomaty, Madagascar	Protect rare and endangered plants and animals	Provides important sources of essential materials (food, firewood, medicinal plants, etc.). The shores of the lakes are also used for agriculture

5. CONCLUSION AND RECOMMENDATIONS

If we want a strong Africa in the future we must lay its foundations on our indigenous knowledge in all areas of our lives. We will borrow ideas and skills from others since we live in an interdependent world but if we are in possession of our minds what we borrow will come to enrich and embellish what we already have and not supplant it"

Opoku, 1999

In the preceding chapters we have read that our continent-Africa is prone to many forms of natural hazards which, if not recognised, will frustrate efforts to improve the well being of our communities through sustainable development.

Although this book is intended to stimulate awareness amongst school-goers and at community level, there are already several major initiatives in place at the international and national level. For further research purposes, a brief outline of the international and national initiatives being undertaken in Africa is given in appendix A, while appendix B briefly summarises partnerships, aid and funding opportunities.

There is much that schools goers can do to bring about a greater awareness of the need for environmental protection and disaster risk reduction. They can support local initiatives and ask questions that will generate discussion amongst their elders and stimulate their community's interest in the subject.

5.1. A SUGGESTION!

Below are examples of the kinds of questions that can be asked to determine how best your community is implementing environmental protection and disaster risk reduction measures in your area. Why not use them to generate classroom discussions and debates? Or form the subject for a school research project?

Knowledge Development

1. Is awareness about the importance of disaster risk reduction and environmental protection taught in your school and to the adult education groups in your community?

2. Find out if there are any local initiatives to determine which the local areas at risk are, and which require environmental protection.
3. Can you identify potential support and partners from nearby schools, neighbouring areas and the regional authorities such as your regional or district education authority?
4. Could you initiate a class project to produce posters and other informational material which would help inform your community about disaster reduction in your area?

Risk and Environmental Assessment

5. Is your area at risk from any of the following: earthquakes, volcanoes, landslides, rock falls and surface collapse? This, and the suggestions below, could be the subjects of geography research projects.
6. Is your area at risk from any of the following floods, tropical cyclones, hailstorms, drought, wildfires and sandstorms?
7. Is your area at risk from any of the following: human, animal or plant diseases and infestations of alien vegetation?
8. Is your community at risk from any of the following: industrial pollution, dam failure, toxic waste or technological accidents?
9. Are any of the following processes active in your area? Land degradation, desertification, deforestation, loss of biodiversity, land, water and air pollution?
10. Can you map-out the areas surrounding you that are most at risk?
11. What would be the worst rapid-onset disaster that could strike your community?
12. What slow-onset disaster is your community at the greatest risk from?
13. Can you identify any local areas which are at the most risk?
14. What steps could be taken to reduce their risk?
15. Identify which activities cause the greatest damage to your local environment?
16. What could be done to reduce the negative impact of these activities on the environment?
17. Are these being implemented? If not, why not?

Awareness

18. Find out what information is locally available that informs the community about living in potentially dangerous areas.
19. What are the consequences of the unsustainable use of natural resources?
20. What are the consequences of the environmental degradation?
21. Are undesirable land use practices discussed with the people involved?
22. Are they actively encouraged to change their land use practices where necessary?

Early Warning Systems

23. If potential hazards have been identified that could cause damage to your community and the surrounding environment, try and find out what steps are being taken to monitor and measure changes so that your community can have early warning of such potential disasters.
24. How would your local authority go about warning your community that a potential disaster is imminent?
25. Does your school participate in measuring environmental degradation, because this could be an important indicator of a slow-onset disaster?

Integrated Environmental and Disaster Risk Management Plans

As a future adult and a responsible member of your community, you need to understand how it is organised and who is responsible for what.

26. Which local organisation is responsible for the application of appropriate risk reduction and environmental impact mitigation measures in your area?
27. Does your community have a Disaster Management Centre? Or at least an official who is responsible for disaster management?
28. Have response plans been prepared in the event of an unavoidable disaster?
29. Do you and your family know what to do and where to go in the event of an emergency in your community?

5.2. EPILOGUE

As school goes you can be part of a worldwide awakening of the need to better protect our environment. We trust that this booklet has assisted you in gaining some insight into this global problem and what we in Africa can do to advance environmental protection and sound risk management. **After all tomorrow is in your hands!**

African communities that can adapt and learn from disaster risk reduction and environmental protection principles will not only be securing a more sustainable future for themselves, their children and their grandchildren, but will also be an inspiration to others of Africa's emerging self reliance.

APPENDIX A

INTERNATIONAL AGREEMENTS

International Agreements and Commitments

Environmental Protection and Disaster Risk Reduction are recognised as an essential part of sustainable development. At the UN Conference on Human Environment (Stockholm, 1972) the international community met for the first time to consider the global environment and highlighted the need to support people in this process.

The UN Conference on Environment and Development, also known as the Earth Summit, took place in 1992, in Rio de Janeiro. The main outputs of that Summit were the Rio Declaration on Environment and Development and Agenda 21 (a programme of action).

Disaster Reduction was not specifically dealt with in this declaration but during the International Decade for Natural Disaster Reduction; 1989-1999 the connection between disaster reduction and sustainable development, has been clarified to a greater extent.

In 2002 the World Summit on Sustainable Development held in Johannesburg resulted in:

- A political declaration on increased commitment for action;
- A Plan of Implementation ;
- The formation of partnerships to implement the Plan of Implementation.

AFRICAN/SUB-REGIONAL INITIATIVES

Many African countries are party to various environmental policy instruments at regional and sub-regional level. The formation of the African Union and NEPAD provides an opportunity to place Africa on a path of sustainable growth and development by reducing poverty, revitalising governance and developing mutually beneficial global partnerships.

A number of disaster reduction initiatives between African Institutions and the UN ISDR Africa are in progress. These include:

- Sub-regional and regional reviews;

- Development of guidelines to mainstream Disaster Risk Reduction into development projects;
- Promoting women's active participation in disaster reduction;
- The development of educational materials for children and community leaders;
- The development of an African Regional Strategy on Disaster Reduction together with the African Union and NEPAD.

KEY INTERNATIONAL EVENTS RELATED TO DISASTER RISK REDUCTION AND SUSTAINABLE DEVELOPMENT

- 1972 - UN Conference on Human Environment (Stockholm) UNEP established
- 1983 - World Commission on Environment and Development (the Brundtland Commission) established
- 1989 - Launch of the International Decade for Natural Disaster Reduction (IDNDR)
- 1992 - UN Conference on Environment and Development (the Earth Summit) in Rio de Janeiro
- 1994 - World Conference on Natural Disaster Reduction, Yokohama
- 1994 - UN Conference on Population and Development
- 1995 - UN Fourth World Conference on Women, Beijing
- 1997 - First World Water Forum, Marrakech
- 1999 - Completion of the International Decade for Natural Disaster Reduction
- 2000- Launch of the International Strategy for Disaster Reduction (ISDR)
- 2000 - Millennium Summit, New York
- 2002 - World Summit on Sustainable Development, Johannesburg
- 2005 - Proposed World Conference on Disaster Reduction, Kobe

APPENDIX B

PARTNERSHIPS, AID AND FUNDING FOR DISASTER RISK REDUCTION

The UN/ISDR encourages networking and partnerships in order to build links that help to reduce risk in Africa. These partnerships can range from information exchange networks to fully organised and funded implementation partnerships. Examples of these are:

- Technical and research networks with governments and universities;
- Rural networking support - involving NGOs working with vulnerable communities (e.g. FAO, USAID, CARE);
- Networking support for community partnerships - consisting of groups of organisations and people committed to disaster risk reduction in communities. In Africa, Periperi is a network of partners from five Southern African countries. Periperi was established by the Disaster Mitigation for Sustainable Livelihoods Programme (DiMP) at the University of Cape Town;
- Cross-sectoral coordination and collaboration - where different government departments and communities all work together to reduce the risk of disasters or manage the impacts of disasters.

GLOSSARY

The following terms are taken from Living with Risk, A global review of disaster reduction initiatives UN/ISDR 2002

Acceptable risk: The level of loss a society or community considers acceptable given existing social, economic, political, cultural and technical conditions.

Capacity: A combination of all the strengths and resources available within a community or organisation that can reduce the level of risk, or the effects of a disaster.

Capacity building: Efforts aimed to develop human skills within a community, organisation or institution needed to reduce the level of risk.

Coping capabilities/Capacity: The manner in which people and organisations use existing resources to achieve various beneficial ends during unusual, abnormal, and adverse conditions of a disaster event or process.

Disaster: A serious disruption of the functioning of a community or a society causing widespread human, material, economic or environmental losses which exceed the ability of the affected community/society to cope using its own resources.

Disaster risk reduction: (disaster reduction) The systematic development and application of policies, strategies and practices to minimise vulnerabilities and disaster risks throughout a society, to avoid (prevention) or to limit (mitigation and preparedness) adverse impact of hazards, within the broad context of sustainable development.

Early warning: The provision of timely and effective information, through identified institutions, that allow individuals at risk of a disaster, to take action to avoid or reduce their risk and prepare for effective response.

Ecosystem: A system of interacting living organisms together with their physical environment.

Environmental degradation: Processes induced by human behaviour and activities (sometimes combined with natural hazards), that damage the natural resource base or adversely alter natural processes or ecosystems. Potential effects are varied and may contribute to increased vulnerability and the frequency and intensity of natural hazards.

Hazard: A potentially damaging physical event, phenomenon or human activity, which may cause the loss of life or injury, property damage, social and economic disruption or environmental degradation.

Hazard analysis: Identification, studies and monitoring of any hazard to determinate its potentiality, origin, characteristics and behaviour.

Land-use planning: Branch of physical planning that determines the most desirable way land should be used. Involves land-use studies and mapping, analysis of data acquired, formulation of alternative land-use decisions and design of a long-range land-use plan for different geographical and administrative scales.

Mitigation: Structural and non-structural measures undertaken to limit the adverse impact of natural hazards, environmental degradation and technological hazards.

Preparedness: Activities and measures taken in advance to ensure effective response to the impact of disasters, including the issuance of timely and effective early warnings and the temporary removal of people and property from a threatened location.

Prevention: Activities to provide outright avoidance of the adverse impact of hazards and related environmental, technological and biological disasters.

Public awareness: The processes of informing the general population, increasing their levels of consciousness about risks and how to take action to reduce their exposure to hazards. This is particularly important for public officials in fulfilling their responsibilities to save lives and property in the event of a disaster.

Recovery: Decisions and actions taken after a disaster with a view to restoring the living conditions of the stricken community, while encouraging and facilitating necessary adjustments to reduce disaster risk.

Relief/response: The provision of assistance and/or intervention during or immediately after a disaster to meet the life preservation and basic subsistence needs of those people affected. It can be of an immediate, short-term, or protracted duration.

Resilience/resilient: The capacity of a system, community or society to resist or to change in order that it may obtain an acceptable level in functioning and structure. This is determined by the degree to which the social system is capable of organizing itself, and the ability to increase its capacity for learning and adaptation, including the capacity to recover from a disaster.

Risk: The probability of harmful consequences, or expected loss (of lives, people injured, property, livelihoods, economic activity disrupted or environment damaged) resulting from interactions between natural or human induced hazards and vulnerable/capable conditions. Conventionally risk is expressed by the equation $\text{Risk} = \text{Hazards} \times \text{Vulnerability} / \text{Capacity}$

Risk assessment/analysis: A process to determine the nature and extent of risk by analysing potential hazards and evaluating existing conditions of vulnerability/capacity that could pose a potential threat or harm to people, property, livelihoods and the environment on which they depend.

Risk management: The systematic management of administrative decisions, organization, operational skills and responsibilities to apply policies, strategies and practices for disaster risk reduction.

Risk reduction measures: The development and application of policies, procedures and capacities of the society and communities to lessen the negative impacts of a possible impact of natural hazards and related environmental and technological disasters. This includes structural and non-structural measures to avoid (prevention) or to limit (mitigation and preparedness) adverse impact of hazards, as well as the development of coping capabilities.

Sustainable development: Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It contains within it two key concepts: the concept of "needs", in particular the essential needs of the world's poor, to which overriding priority should be given; and the idea of limitations imposed by the state of technology and social organization

on the environment's ability to meet present and the future needs. (Brundtland Commission, 1987).

Vulnerability: A set of conditions and processes resulting from physical, social, economical and environmental factors, which increase the susceptibility of a community to the impact of hazards.

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<http://www.etcgroup.org>

Famine Early Warning Network <http://www.fews.net>

Institute of Development Studies <http://www.ids.ac.uk>

International Development Research Centre <http://web.idrc.ca>

International Institute for Environment and Development

<http://www.iied.org>

Intergovernmental Panel on Climate Change <http://www.ipcc.ch>

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ProVention Consortium <http://www.proventionconsortium.org>

United Nations Joint programme on HIV/AIDS <http://unaids.org>

United Nations Convention to Combat Desertification

<http://www.unccd.int/main.php>

United Nations Environment Programme <http://www.unep.org>

United Nations Food & Agricultural Organisation

<http://www.fao.org/publishing>

National Oceanic & Atmospheric Administration

<http://www.photolib.noaa.gov/>

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SADC Food Security Programme, Food, Agriculture and Natural

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International Strategy



The United Nations Secretariat of the International Strategy for Disaster Reduction

Within the United Nations system, the Secretariat of the International Strategy for Disaster Reduction is responsible for Co-ordinating disaster reduction strategies and programmes.

It's mission is to help people withstand disasters by making them aware of the importance of disaster reduction measures and providing support to help reduce human, economic and social losses.

The Secretariat also provides backing for an Inter-Agency Task Force on Disaster Reduction headed by the Under-Secretary-General for Humanitarian Affairs and comprising representatives of several United Nations agencies, regional institutions and non-governmental organisations.

Within the United Nations system, the Task Force is the chief body responsible for the design of disaster reduction policy.

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